Crop Recommendation and Crop Disease Detection using Machine Learning

Domain: Agriculture | IBM University Engagement — AICTE ML Project Submission

Problem Statement

Indian agriculture is highly dependent on selecting the right crop for a given field's soil and climate conditions, and on detecting crop diseases early enough to act. Farmers often rely on experience or generic advice rather than data-driven analysis of soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall — leading to suboptimal yields. Separately, crop diseases such as leaf blight and leaf rust are frequently identified late, by visual inspection alone, causing avoidable yield loss. The task is to design, build, and evaluate (1) a multi-class classification model that recommends the most suitable crop for a given set of soil and climate parameters, and (2) an image-based classification model that detects whether a crop leaf is healthy or affected by a specific disease. Together these form a decision-support tool that can help farmers and agricultural extension workers make faster, more accurate decisions.

Proposed Solution (IBM watsonx.ai Studio – AutoAI)

We propose an AI-powered Crop Advisory System using IBM watsonx.ai Studio and AutoAI. Soil and climate data (N, P, K, temperature, humidity, pH, rainfall) is uploaded to watsonx.ai, where AutoAI automatically performs data preprocessing, feature engineering, algorithm selection, and hyperparameter optimization across multiple candidate pipelines (Logistic Regression, KNN, Decision Tree, Random Forest, Gradient Boosting, SVM). The best-performing pipeline is selected from the leaderboard and further tuned. In parallel, a crop-disease detection module extracts visual features from leaf images and classifies them as Healthy, Blight, or Rust, giving farmers an early warning before a disease spreads.

Technology Used

- IBM Cloud / IBM watsonx.ai Studio — AutoAI-based model search and deployment infrastructure
- Python, scikit-learn — Random Forest, Gradient Boosting, SVM, Logistic Regression, KNN, Decision Tree
- Pandas / NumPy — data preprocessing and feature engineering
- Matplotlib / Seaborn — model evaluation visualizations
- Pillow — image processing for the disease detection pipeline

Results Summary

- Crop Recommendation: Random Forest (tuned via GridSearchCV) — ~95% test accuracy across 12 crop classes
- Crop Disease Detection: Random Forest (image-feature based) — ~89% test accuracy across 3 classes (Healthy, Blight, Rust)

Novelty and Uniqueness

- Combines soil/climate-based crop recommendation with image-based disease detection in a single advisory pipeline.
- Uses an AutoAI-style multi-pipeline leaderboard, automatically discovering the best model without extensive manual coding.
- Automated feature engineering surfaces hidden patterns in socio-agronomic data.
- A consistent, unbiased decision-support layer that reduces reliance on ad-hoc farmer experience alone.

Future Scope

- Integrate with government agricultural portals and Kisan Call Centers for real-time advisory delivery.
- Expand the disease-detection module with a full CNN trained on field-collected image datasets.
- Add real-time weather-API integration for dynamic, season-aware recommendations.
- Scale to state/national level with regional crop-calendar and market-price overlays.